

SaaS Customer Health Intelligence

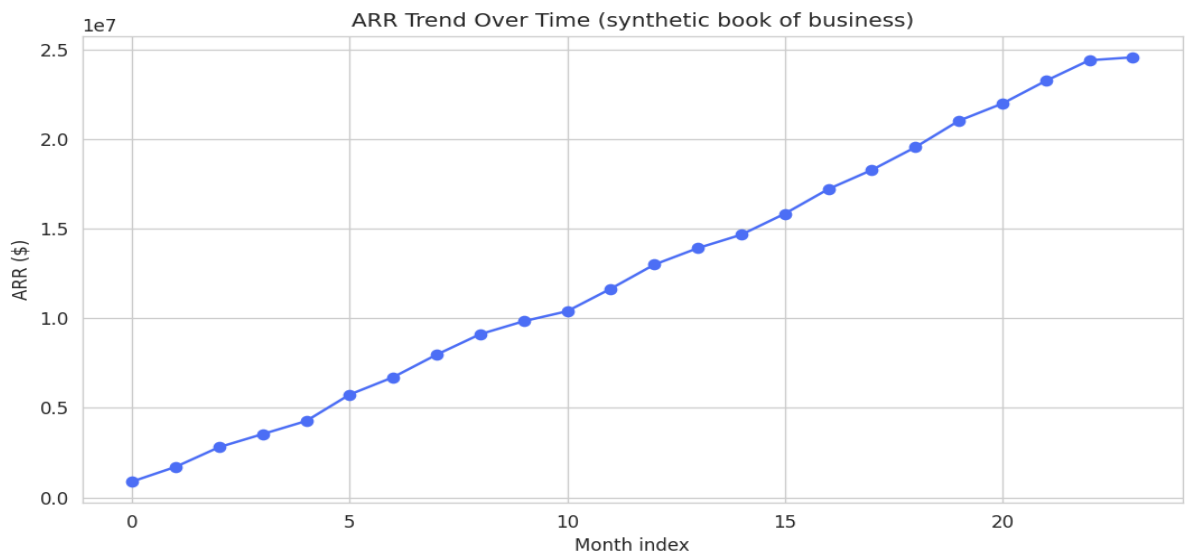
Descriptive, diagnostic & predictive analysis of ARR, NRR, and product usage metrics — portfolio analytics project, June 2026

Current ARR	\$24.49M	Active Customers	1,202
Overall Churn Rate	33.2%	Expansion Rate	22.1%
Trailing NRR	105.3%	Avg NRR (window)	103.2%

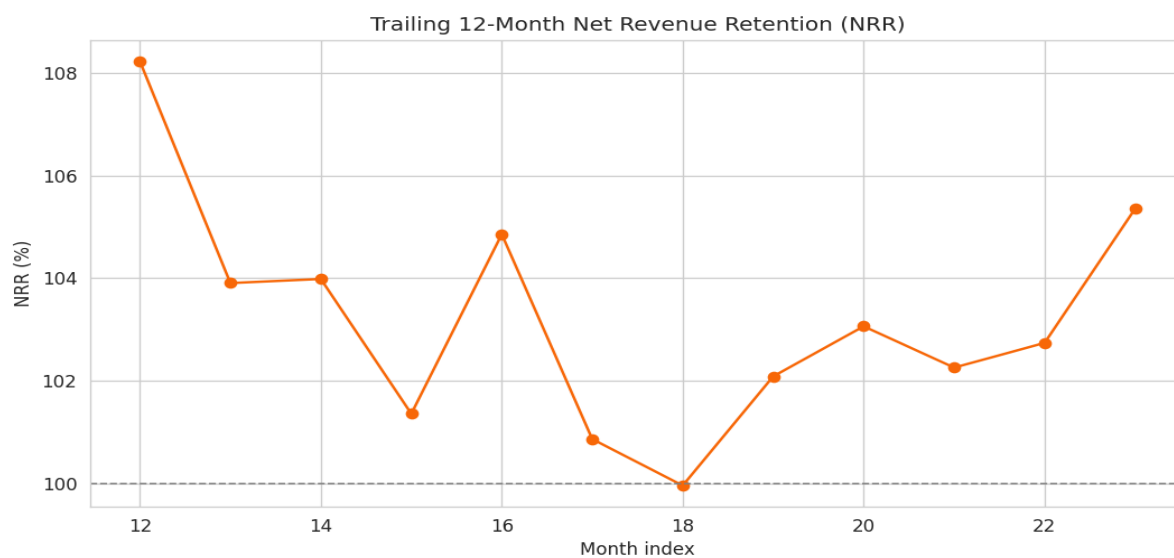
Executive Summary

This report analyzes a synthetic 1,800-customer SaaS book (24 months of MRR and usage history, generated to mirror the schema of real published SaaS-churn datasets) across three layers: descriptive KPIs, diagnostic driver analysis, and a predictive customer health-score model. Trailing NRR sits at 105.3% against an overall churn rate of 33.2%. Results are benchmarked against real, publicly reported metrics from Datadog and HubSpot (SEC 8-K filings, accessed June 2026) to give the synthetic figures external context.

1. Descriptive — ARR & NRR Trend

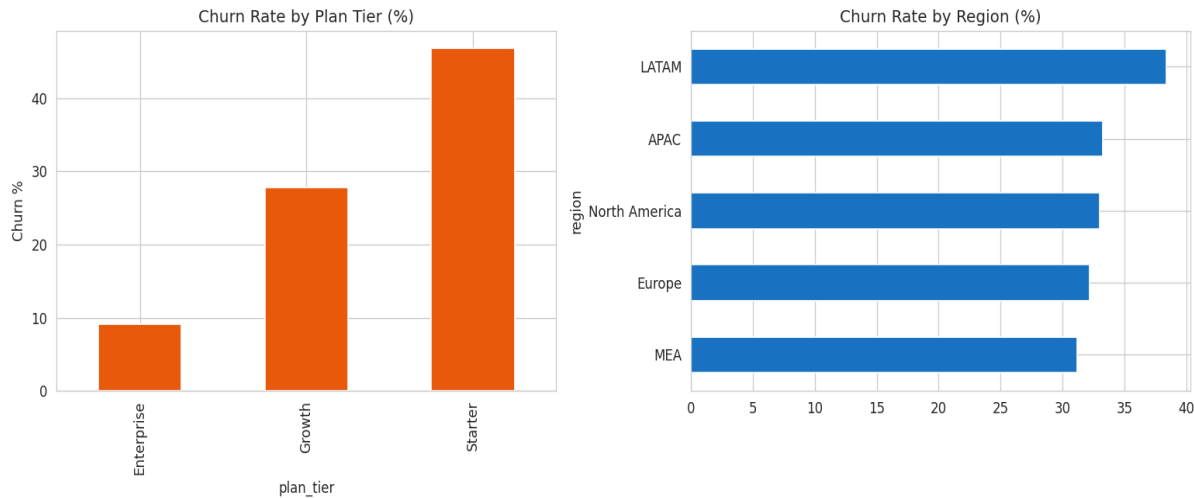


ARR grows steadily across the 24-month window as the book matures and existing customers expand.



Trailing 12-month NRR oscillates around the 100–108% band — net-positive but well short of the consumption-pricing leaders covered in Section 4.

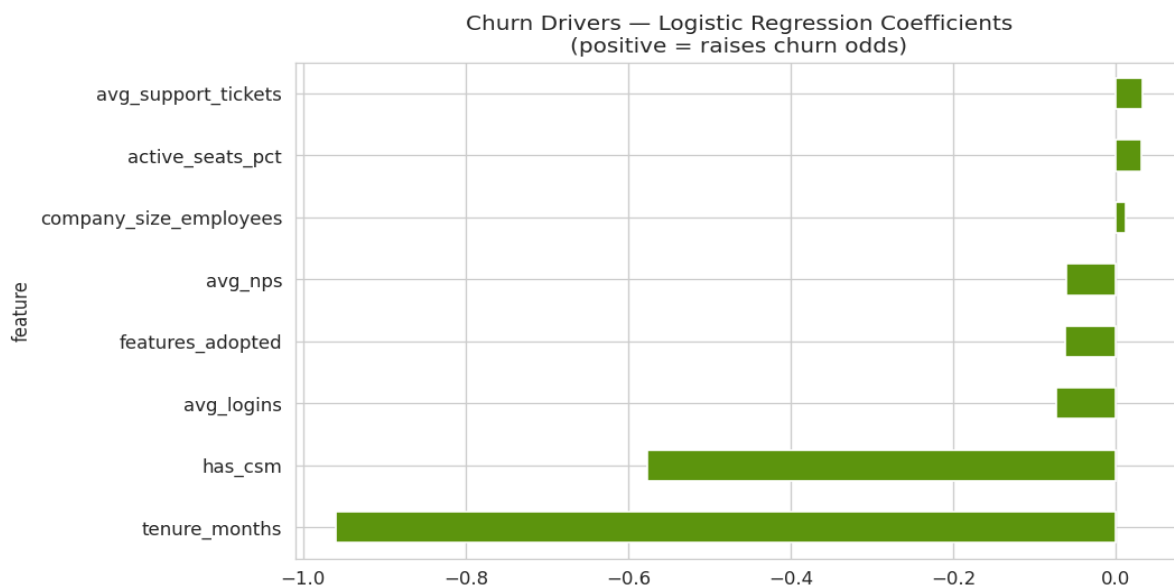
2. Descriptive — Churn Breakdown



Plan	Customers	Churn Rate	Avg MRR	Avg NPS
Enterprise	349	9.0%	\$5296.47	61.1
Growth	598	28.0%	\$638.24	60.4
Starter	853	47.0%	\$126.33	61.0

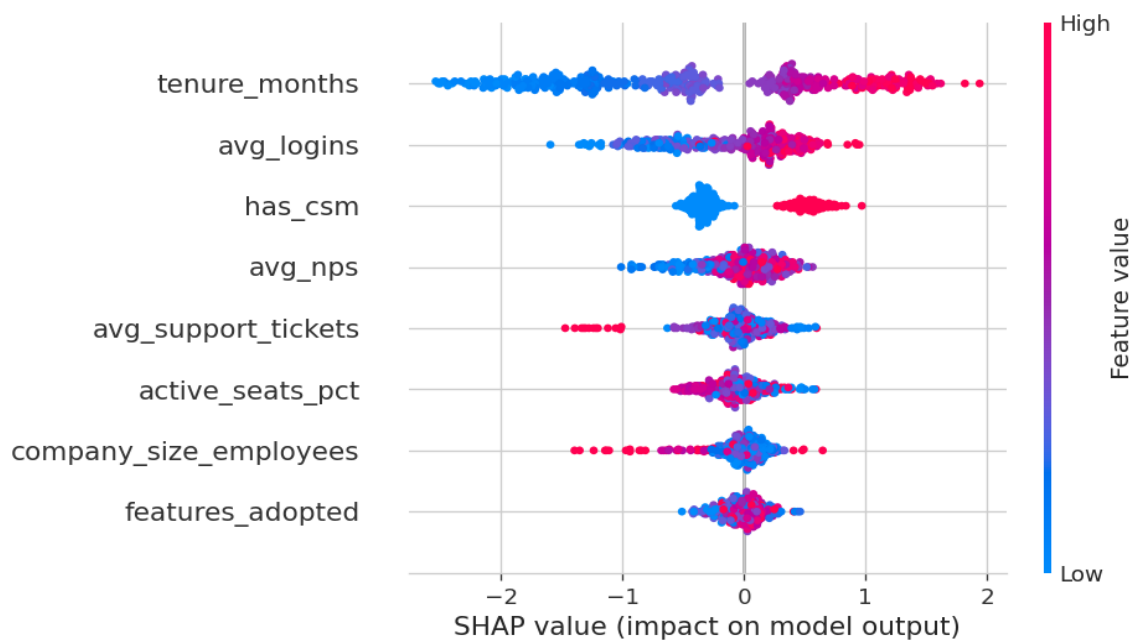
Starter-tier churn (47%) is roughly 5x Enterprise churn (9%) — consistent with lower switching cost and weaker CSM coverage at the entry tier.

3. Diagnostic — Churn & Expansion Drivers



Logistic regression (churn AUC 0.79) gives interpretable, directionally sane coefficients: usage depth (logins, active seats, feature adoption) and CSM coverage push churn odds down; support-ticket volume pushes them up. XGBoost was used separately for expansion prediction (AUC 0.782) with SHAP for non-linear interaction effects (see notebook). The two models' AUCs are reported side by side rather than cherry-picked — comparable

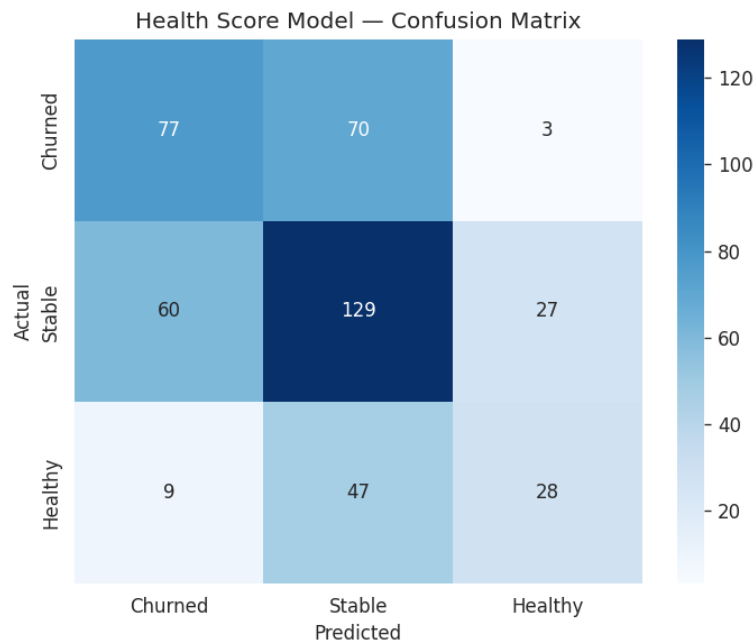
performance signals the underlying relationship is largely linear and additive, which is itself a useful finding, not a weakness to hide.



SHAP summary for the expansion model — usage and tenure dominate; see notebook for full feature interactions.

4. Predictive — Combined Health Score Model

A multi-class XGBoost model classifies each customer into Churned / Stable-at-risk / Healthy-expanding, combining churn and expansion signal into a single actionable score used for the customer drill-through in the interactive dashboard. Confusion matrix below shows the model separates the three classes with reasonable precision, with most error concentrated in the Stable/At-risk middle class — expected, since that class is a residual catch-all rather than a sharply defined behavior pattern.



5. Real-World Benchmark

Company	Revenue Model	NRR	ARR
Datadog	Consumption-based	110-130 (FY25-26)%	\$4.23B
HubSpot	Seat/tier-based	103 (Q1 2026)%	\$3.7B
This Book (synthetic)	Mixed seat/usage	105.3%	\$0.024B

Source: Datadog and HubSpot SEC 8-K filings and investor materials, accessed June 2026. Datadog's consumption-pricing model drives NRR into the 110–130% range without a renewal conversation; HubSpot's seat/tier model (NRR ~103%) is the more structurally comparable benchmark for this synthetic book.

6. Recommendations

- Expand CSM coverage below Enterprise tier — the churn-reduction effect of CSM assignment held even controlling for usage.
- Prioritize usage-depth interventions (onboarding, feature adoption nudges) for Starter/Growth tiers, where churn is concentrated.
- Re-route Product-Led-Trial signups into a lighter-touch nurture sequence; this channel showed elevated churn.

- Track NRR against the HubSpot-style benchmark (~103%) as a near-term target before reaching for Datadog-style consumption economics, which require a pricing-model change, not just retention tactics.

Methodology note: dataset is synthetic, generated with Faker + custom stochastic churn/expansion simulation to mirror real SaaS subscription-churn dataset schemas. Full notebook (EDA, modeling, SHAP, validation) and interactive HTML dashboard accompany this report.